

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE					
Supplementary Winter Examination – 2024					
Course: B.Tech Branch : Electrical Engineering / Electrical Engg (Electronics & Power)/Electrical & Electronics Engg/Electrical & Power Engg					
Semester : VII					
Subject Code & Name: BTEEE704B Electrical Traction and Utilization					
Max Marks: 60		Date: 12/02/2025		Duration: 3 Hr	
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.					
			(Level/CO)	Marks	
Q.1	Objective type questions. (Compulsory Question)			12	
1	In direct arc furnace which of the following has highest value?			Understand	1
	a. Current	b. Voltage	c. Power factor		
2	 Is used for heating non-conducting materials			Understand	1
	a. Eddy current heating	b. Arc heating	c. Induction heating		
3	The illumination at the various points on a horizontal surface illuminated by the same source varies as				1
	a. $\cos \theta$	b. $\cos^2 \theta$	c. $\cos^3 \theta$		
4	due to Moonlight illumination is nearly				1
	a. 300 lumen/m ²	b. 30 lumen/metres ²	c. 0.3 lumen/m ²		
5	Mercury vapour lamp gives light of				1
	a. Green colour light	b. Pink colour light	c. Greenish blue colour light		
6	Luminous flux is			Understand	1
	a. The rate of energy radiation	b. The part of light energy	c. Measured in lux		

	in the form of light waves	radiated by sun that is received on earth				
7	Projection welding is.....					1
	a. Multi-spot welding process	b. Continuous spot welding process	c. Used to form mesh	d. Used to make cantilevers		
8	In underground traction the supply system is					1
	a. 500 V to 2000 V DC	b. 25KV, 50 HZ	c. 500 KV, 50 HZ	d. 25 KV, 25 HZ		
9	Low-frequency operation of overhead line in traction system					1
	a. Increases the spacing between substations	b. Reduces the spacing between substations	c. Spacing is independent of frequency of supply	d. All of the above		
10	Quadrilateral speed time curve pertains to which of the following services ?					1
	a. Main line service	b. Urban service	c. Sub Urban service	d. Urban and suburban service		
11	Longer coasting period for a train results in				Understand	1
	a. Higher schedule speed	b. Lower specific energy consumption	c. Higher retardation	d. Higher acceleration		
12	Which of the following is an advantage of electric traction over other methods of traction ?				Understand	1
	a. Faster acceleration	b. No pollution problems	c. Better braking action	d. All of the above		
Q.2 Solve the following.						12
A)	Explain resistance heating, list out the advantages of electrical heating.				Remember	6
B)	Explain in detail any one method of welding.				Understand	6
Q.3 Solve the following.						12
A)	State faradays laws of electrolysis & Explain application of electrolysis.				Understand	6
B)	Write a short note on Accumulators & cell.				Remember	6

Q.4	Solve Any Two of the following.		12
A)	Design of illumination scheme-Factors to be considered for design of illumination scheme.	Understand	6
B)	Calculation of illumination at different points, considerations involved in simple design problems for indoor installation	Remember	6
C)	Explain factors to be considered for design of indoor & outdoor lighting.	Understand	6
Q.5	Solve Any Two of the following.		12
A)	Describe design & Procedure for factory lighting.	Remember	6
B)	Explain Characteristics of traction motors.	Understand	6
C)	Explain the requirement of ideal traction system and also gives advantages and disadvantages.	Remember	6
Q.6	Solve Any Two of the following.		12
A)	Explain speed time curves of main line service	Remember	6
B)	Explain Tractive effort to overcome train resistance	Understand	6
C)	Classify mechanical verses electrical braking.	Remember	6
	*** End ***		

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